COMPREHENSIVE REPORT ON PEAK HOUR TRAFFIC ANALYSIS

Name: Tejavath Shalini
Project: Traffic Volume Prediction at Road Junctions
Menternship: Uber Traffic Analysis Project

INTRODUCTION:

This report analyzes traffic congestion patterns at different road junctions using historical traffic data. The objective of this study is to identify peak traffic hours and understand how external factors such as weather conditions and special events influence traffic congestion. The insights obtained from this analysis can help improve traffic management strategies and reduce congestion during peak periods.

DATASET DESCRIPTION:

The dataset includes traffic volume data collected at multiple junctions along with weather conditions

and information about special events. The main variables in the dataset include:

- Date and Time
- Junction ID
- Traffic Volume
- Temperature
- Rain indicator
- Humidity
- Special events (Festival, Holiday, Marathon, Public events)

These variables help analyze how traffic patterns change across time and conditions.

METHODOLOGY:

The analysis involved the following steps:

- Data preprocessing and feature engineering

• Calculation of congestion metrics using average traffic volume per hour
• Identification of peak traffic hours for each junction
• Analysis of temporal patterns including weekday, weekend, and monthly variations
• Examination of the impact of weather and special events on traffic congestion
• Visualization of results using graphs and heatmaps

KEY FINDINGS:

The analysis revealed several important insights:

• Peak traffic hours occur between 8–10 AM and 5–7 PM.

• Junction J2 experiences the highest congestion levels.

• Traffic volume is generally higher on weekdays compared to weekends.

• Weather conditions such as rain contribute to increased congestion.

• Special events such as festivals and marathons lead to temporary spikes in traffic.

RECOMMENDATIONS:

Based on the analysis, the following recommendations are proposed:

- Implement adaptive traffic signal control during peak hours.
- Encourage commuters to travel during off-peak periods.
- Increase monitoring and traffic management during special events.
- Promote public transportation during high congestion periods.
- Use predictive models to forecast traffic congestion and plan mitigation strategies.